

Faculty Vitae: Sangil Kim

1. Education

B.S. Polymer Science and Engineering, Chungnam National University South Korea, 1997
M.S. Polymer Science and Engineering, Chungnam National University South Korea, 1999
M.S. Chemical Engineering, University of Cincinnati, Cincinnati, OH, 2003
PhD Chemical Engineering, Virginia Tech, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, Associate Professor,
2022 – present, Full-Time
Department of Chemical Engineering, University of Illinois, Chicago, Assistant Professor,
2015 – 2022, Full-Time.

3. Non-Academic Experience

Lawrence Livermore National Lab, Livermore, CA, Guest Researcher, 2015 – present, Part-Time
Lawrence Livermore National Lab., Livermore, CA, Staff Scientist, 2011– 2015, Full-Time
Porifera Inc, Hayward, CA, Research Scientist, 2009– 2011, Full-Time
University of California Davis/Lawrence Livermore National Lab., Livermore, CA,
Post-doctoral Associate, 2007 – 2009, Full-Time
Korea Institute of Science and Technology (KIST), Seoul, South Korea, Researcher, Complex
Fluid and Membrane Laboratory, 2001, Full-Time

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

Electrochemical Society, Materials Research Society, American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

- 2021-present: Editorial Board Member, *Membranes*
- Reviewer for journals (e.g., Langmuir, Journal of Membrane Science, Nanotechnology, Journal of Colloid and Interface Science, Journal of Microfluidics and Nanofluidics, Journal of Physics:Condensed Matter)
- Proposal reviewer for DOE-EERE and NSF

8. Selected Publications

Patents

- 1) J Ye, AC Baumgaertel, J Biener, MM Biener, S Kim, “Three dimensional nanoporous membrane having multiple independent, continuous pore systems”, U.S. Patent 11,103,832, 2021
- 2) M. Worley, P. Campbell, S. Kim, and M. Merrill, "Molten Hydroxide Membrane for Separation of Acid Gases from Emissions", U.S. Patent 10,464,015, 2019.

- 3) PD. Hoeprich and S. Kim, “Electrochemical flow-cell for hydrogen production and nicotinamide dependent target reduction, and related methods and systems”, U.S. Patent 10,151,037, 2018.
- 4) S. Kim, P. Naraghi-Arani, L. Megan, “Filtration Device for Rapid Separation of Biological Particles from Complex Matrices”, U.S. Patent 9,861,939, 2018.
- 5) M. Worsley, J. Satcher, S. Kucheyev, S. Charnvanichborikarn, J. Colvin, T. Felter, S. Kim, M. Merrill, C. Orme, “Nanoporous Metal-Carbon Composites”, U.S. Patent 9,844,762, 2018.
- 6) R. Revanur, V. Lulevich, I. Roh, J. Klare, S. Kim, A. Noy, and O. Bakajin, “Membranes having aligned 1-D nanoparticles in a matrix layer for improved fluid separation”, U.S. Patent, 9,216,391, 2015.
- 7) O. Bakajin, A. Noy, F. Fornasiero, H. Park, J. Holt and S. Kim, “Membranes with Functionalized Carbon Nanotube Pores for Selective Transport”, U.S. Patent 8,940,173, 2015.
- 8) E. Marand and S. Kim, “Method for Making Oriented Carbon Nanotube/Polymer Nano-Composite Membranes”, U.S. Patent 7,931,838, 2011.

Book chapters

- 1) O. Bakajin, A. Noy, F. Fornasiero, C. P. Grigoropoulos, J. K. Holt, J. In, S. Kim, H. Park, Nanofluidic Carbon Nanotube Membranes: Applications for Water Purification and Desalination, Nanotechnology for Water Quality, Eds: N. Savage, et al., William Andrew Publishers, 2008.

Peer-reviewed publications in last 5 years (*co-corresponding author)

- 1) K. Wang, N. Hwee, W. Degraff, S. M. Biener, S. Huang, L. Feng, S. Watts, W. L. Smith, B. Wang, S. Chatterjee, T. W. Heo, J. Ye, G. Cheng, J. Biener, S. Kim*, “Zwitterionic Photocurable Resin for High-Resolution 3D Printing of Ultralow-Fouling Microstructures”, *Small Methods*, e01222, 2025
- 2) KM. Lwin, T. Kim, J. Lee, S. Han, J. Bae, S. Park, S. Kim*, K. Kim*, “Cost-effective thin metal–polyphenol coating for enhancing fouling resistance of ultrafiltration membranes”, *Sustainable Materials and Technologies*, e01509, 2025
- 3) K. Wang, V. Koverga, N. Maslekar, F. Wu, R. Kuphl, X. Lyu, P. Deshpande, H. Guo, H. Seol, W. Degraff, J. L. Schaefer, C. Fang, T. Li, G. Cheng, A. T. Ngo, S. Kim, “Novel Zwitterionic Polyurethane-in-Salt Electrolytes with High Ion Conductivity, Elasticity, and Adhesion for High-Performance Solid-State Lithium Metal Batteries,” *Advanced Energy Materials*, 2405676, 2025
- 4) L. Feng, S. Huang, D. Li, K. Wang, S. Kim, TW. Heo, J. Biener, “Holistic Microstructure Control Strategies in Photopolymerization-Induced Phase Separation of Acrylate Systems”, *Small Structures*, 2500081, 2025
- 5) S. Misal, D. Li, S. Kim*, B. Chaplin*, “Effect of Solution Conditions and Applied Potential on Ion Transport in TiO₂ Nanopores”, *ACS ES&T Engineering*, 2495-2505, 2024
- 6) S. Cetindag, A. Pendse, R. J. Castellano, P. Rehak, M. Howard, K. Wang, J. Yi, R. Praino, P. Kral, L. Liu, S. Kim*, J. Shan*, “Anomalous Diffusive and Electric-field-driven Ion Transport in Boron-Nitride Nanotubes”, *ChemRxiv*, 2023 (currently 2nd revision at *Nature Nanotechnology*)
- 7) K. Wang, H. Seol, A. Cheng, N. McKeague, M. Carlson, W. Degraff, S. Huang, and S. Kim, “Simple Bioparticle Filtration Device Based on an Ultralow-Fouling Zwitterionic

- Polyurethane Membrane for Rapid Large-Volume Separation of Plasma and Viruses from Whole Blood”, *Membranes*, 13, 524, 2023
- 8) A. Pendse, S. Cetindag, K. Wang, D. Li, R. J. Castellano, T. Wang, J. W. Shan, and S. Kim, “Intrinsic Ion Transports of Highly Charged Sub-3-nm Boron Nitride Nanotube”, *Materials Today*, 60, 79-90, 2022
 - 9) K. Wang, T. Arado, A. Huner, H. Seol, X. Liu, H. Wang, L. Hassan, K. Suresh, S. Kim*, G. Cheng*, “Thermoplastic zwitterionic elastomer with critical antifouling properties”, *Biomaterials Science*, 10, 2892-2906, 2022
 - 10) T. Wang, J. Lee, X. Wang, K. Wang, C. Bae, and S. Kim, “Surface-Engineered Nafion/CNT Nanocomposite Membrane with Improved Voltage Efficiency for Vanadium Redox Flow Batteries”, *Journal of Applied Polymer Science*, 51628, 2021
 - 11) K. Wang, H. Seol, X. Liu, H. Wang, G. Cheng*, and S. Kim*, “Ultralow-Fouling Zwitterionic Polyurethane-Modified Membranes for Rapid Large-Volume Separation of Plasma from Whole Blood”, *Langmuir*, 37, 10115, 2021
 - 12) T. Wang, X. Wang, A. Pendse, Y. Gao, K. Wang, C. Bae*, and S. Kim*, “High-efficient multifunctional electrochemical membrane for lithium polysulfide redox flow batteries”, *Journal of Membrane Science*, 636, 119539, 2021
 - 13) A. Pendse, S. Cetindag, P. Rehak, S. Behura, H. Gao, N.H.L. Nguyen, T. Wang, V. Berry, P. Kral, J. Shan*, and S. Kim*, “Highly Efficient Osmotic Energy Harvesting in Charged Boron-Nitride-Nanopore Membranes”, *Advanced Functional Materials*, 31, 2009586, 2021 (*co-corresponding author)
 - 14) T. Wang, J. Han, K. Kim, J. Lee, A. Muenchinger, Y. K. Choi, KD Kreuer, C. Bae*, and S. Kim*, “Novel Sulfonated Aromatic Polymer Ion Exchange Membranes for Breaking the Conventional Proton Selectivity/Conductivity Trade-off Limitation”, *Materials Advances*, 1, 2206, 2020 (*co-corresponding author)
 - 15) T. Wang, S. Liang, Z. Qi, M. Biener, M. A. Worsley, T. Braun, Y. M. Wang, J. Biener, S. Kim*, and J. Ye*, “Nanoengineered 3D nm-thick Biomimetic Membrane for Ultrafast Selective Mass Transportation”, *Materials Horizons*, 7, 2422 - 2430, 2020 (†co-corresponding author)
 - 16) T. Wang, J. Y. Jeon, J. Han, J. H. Kim, C. Bae*, and S. Kim*, “Poly(terphenylene) Anion Exchange Membranes with High Conductivity and Low Vanadium Permeability for Vanadium Redox Flow Batteries (VRFBs)”, *Journal of Membrane Science*, 598, 117665, 2020 (*co-corresponding author)

9. Professional Development

Not applicable